CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2013
EC312 Digital Signal Processing

Date: 04.05.2013, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Draw Block Diagram of Digital Signal Processing System. Explain each block in brief. **[04]**

(b) Determine if the systems described by the following input-output equations are linear or nonlinear. (a) $y(n) = nx(n)$ (b) $y(n) = x(n^2)$ (c) $y(n) = x^2(n)$ **[03]**

Q - 2 (a) Define and Prove Stability and Causality of Discrete Time Systems. **[07]**

(b) Compute the convolution y(n) and correlation r(n) sequences for the following pair of signals and comment on the results obtained. **[07]**
$x(n)=\{\underset{\uparrow}{1},2,4\}$ and $h(n)=\{\underset{\uparrow}{1},1,1,1,1\}$

OR

Q - 2 (a) Draw the block diagram of Analog to Digital converter. Explain Sampling and Aliasing. **[07]**

(b) The analog signal given below is sampled by 1200 samples per second **[03]**

$$x(t) = 4\sin 960\pi t + 6\sin 1440\pi t$$

Find (i) Nyquist Sampling Rate (ii) Maximum Frequency (iii) Discrete Signal x[n].

(c) Explain Frequency Response of Linear Time Invariant System. **[04]**

Q - 3 (a) Prove Differentiation in Z domain property of Z-Transform. By using this property, Determine Z-Transform and sketch ROC of given sequence. **[07]**

$$x[n] = na^n u(n)$$

(b) Determine Z-transform and sketch the ROC of following sequence. **[03]**

$$x[n] = \left(\frac{-1}{6}\right)^n u(n) - \left(\frac{1}{3}\right)^n u(-n-1)$$

(c) Find Inverse Z-Transform of given X[z] using Partial Fraction Expansion Method. **[04]**

$$X[z] = \frac{z^2}{[z-0.5][z-1]^2}$$

OR

Q - 3 (a) Determine the Regions of Convergence (ROC) of finite or infinite duration right-sided, left sided and two-sided sequences. [07]

(b) Determine Z-transform and sketch the ROC of following sequence. [03]

$$x(n)=(\sin \omega_0 n)u(n)$$

(c) Find Inverse Z-Transform of given X[z] using Residue Method. [04]

$$X[z]=\frac{1}{[z-1][z-0.5]}$$

SECTION – II

Q - 4 (a) Find Step Response and Impulse Response for the system whose transfer function is given as $H[z]=\dfrac{z^3+1}{[z+1][z-1]}$ [03]

(b) Define Digital filter. Write down the advantages of Digital Filter over Analog Filter. [04]

Q - 5 (a) Explain Bilinear Z Transform Method of Infinite Impulse Response Filter Design. [07]

(b) The Transfer function of discrete time causal system is given below. [07]

$$H[z]=\frac{1-z^{-1}}{1-0.2z^{-1}-0.15z^{-2}}$$

(a) Find Difference Equation.

(b) Draw Cascade Form Realization.

(c) Draw Parallel Form Realization.

OR

Q - 5 (a) Explain Different Types of Windows of Finite Impulse Response Filter Design. [07]

(b) Obtain Direct Form-I, Direct Form-II, Cascade Form and Parallel Form Realizations for this system $y(n)=\frac{3}{4}y(n-1)-\frac{1}{8}y(n-2)+x(n)+\frac{1}{3}x(n-1)$. [07]

Q - 6 (a) If X[n] is given as {1,2,2,1}.Find corresponding Discrete Fourier Transform(DFT) of X[k] using Decimation In Time Fast Fourier Transform(DIT FFT) Algorithm. [07]

(b) Define Cyclic Property of Twiddle Factor. Find out 4-point DFT of {1,1,0,0}. [07]

OR

Q - 6 (a) If X[n] is given as {1,2,2,1,0,0,0,0}.By using butterfly structure of Decimation in Frequency (DIF),Calculate output for the each stage. [07]

(b) Perform Circular Convolution of given two sequences. [07]

x[n]={1,3,5,3} and y[n]={2,3,1,1}.

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